MIL-R-60199(MU)
AMENDMENT 5
22 April 1970
SUPERSEDING
AMENDMENT 4
7 October 1969

MILITARY SPECIFICATION

READ AMPLIFIER: 10526589

This amendment forms a part of Military Specification MIL-R-60199(MU), dated 24 February 1965.

Paragraph 3.4.6(e): Change "0.6 ± 0.05 Vdc" to "0.65 ± 0.1 Vdc".

Following paragraph 4.1.1, add paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2.

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 units manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

Paragraph 4.2.3: Add the following sentence: "No lots shall be accepted until control samples have been satisfactority tested."

Paragraph 4.3.2: Delete first sentence and substitute:

"The units shall be selected at random by a Government representative as control samples from each 3 months production or portion thereof as follows:

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3 units per first 100 produced
5 units per 200
7 units per 300
9 units per 400
10 units per 500
12 units per 700
15 units per 1000
and 2 units for each 100 items produced in excess of 1000."

Paragraph 4.4.1.1, Item 11: Delete "equal to or less than 2 ohms from D.C. to 500kc" and substitute "100 mv peak-to-peak, output impedance equal to or less that 2 ohms from D.C. to 500kc".

Paragraph 4.5.1.3.2: Delete in its entirety and substitute:

"4.5.1.3.2 Operating.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that the humidity shall be maintained at 92 to 97 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test period, the board shall be capable of performing the tests of 4.5.2.2 and 4.5.2.7. In order to demonstrate this capability, the tests of 4.5.2.2 and 4.5.2.7 shall be performed a minimum of once each 6 hours of the 24 hour period. The final test shall be conducted during the final portion of the 24 hour period. Between tests, the board shall be electrically energized as described in 4.5.2.2."

Paragraph 4.5.1.4: Delete in its entirety and substitute:

"4.5.1.4 Salt spray.- This test shall be conducted in accordance with the salt spray test defined in Method 101, Condition B of MIL-STD-202 using a 5 percent salt solution. At the end of the exposure period, salt deposits shall be removed from the board by a gentle wash, not exceeding 30 minutes, in running water not warmer than 100°F, using a soft-hair brush or plastic-bristle brush. The board shall then be air dried in a controlled environment at 72°F ± 5°F and a relative humidity of 30 to 50 percent for a period of 24 hours. The board shall be inspected visually for any physical damage, then tested in accordance with 4.5.2.2 and 4.5.2.7."

TABLE V, Line 10: Delete.

Paragraph 4.5.2.3(d), line 3: Delete "less than +6.0 ± 0.5" and substitute "+5.5 ± 0.5".

TABLE IX: Delete in its entirety and substitute new Table IX.

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"TABLE IX

COLUMN A	COLUMN B
TP 5	-2.8 ± 0.6 Vdc
TP 6	-2.6 ± 0.6 Vdc
TP 7	-3.6 ± 1.0 Vdc
TP 8	-3.3 ± 1.0 Vdc
TP 9	-6.0 ± 1.5 Vdc
TP 10	-8.3 ± 2.0 Vdc
TP 11	-13.4 ± 1.5 Vdc
TP 12	-5.6 ± 1.5 Vdc
TP 13	-8.0 ± 2.0 Vdc
TP 14	-13.1 ± 1.5 Vdc
PIN 41B	+0.5 ± 0.5 Vdc
PIN 45B	-12.2 ± 2.0 Vdc
PIN 46B	-12.5 ± 2.0 Vdc
PIN 48B	-3.8 ± 1.0 Vdc
PIN 50B	+0.8 ± 0.5 Vdc"

Paragraph 4.5.2.8(1): Delete "0.6 0.05 Vdc" and substitute "0.65 ± 0.1 Vdc".

* Paragraph 4.5.2.8(p): Delete sub-paragraph (p) in its entirety and substitute new sub-paragraph..

"(p) Connect ground to Pin 8A. Measure the voltage from Pin 6A to Pin 8B. The voltage shall be less than 0.034 Vdc. Remove ground from Pin 8A. Connect ground to pin 5A. Measure the voltage from Pin 6A to Pin 1A. The voltage shall be less than 0.034 Vdc. Remove ground from Pin 5A".

* The outside margin has been marked to indicate where change to the previous amendment has been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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